MS-ML Studio: An open-source MATLAB tool for unified mass-spectrometry preprocessing, feature extraction, and classification

Calebe Elias Ribeiro Brima[a], Marilson Reque[a], Cibelle B. Dallagassa[c], Cyntia M. T. Fadel-Picheth[c], Luciano F. Huergo[b], Roberto Tadeu Raittz[a,*]

[a] *Departamento de Bioinformática, Universidade Federal do Paraná, Brazil*
[b] *Setor Litoral, Universidade Federal do Paraná, Brazil*
[c] *Departamento de Patologia Médica, Universidade Federal do Paraná, Brazil*

Abstract

Although many mass-spectrometry (MS) preprocessing and peak-detection tools exist, few free, open-source options integrate raw-spectral import, discriminant feature learning, and classifier training in one interactive graphical workflow for users without programming expertise. **MS-ML Studio** is an open-source MATLAB application that unifies preprocessing (normalization, reshaping, optional alignment), three complementary, cross-validated feature extractors (correlation-based peak selection, Watch Points, Feature Resonance with genetic-algorithm optimization), and multilayer perceptron (MLP) classification with in-GUI visualization. As a **reference proof-of-concept**—not a clinical validation—we applied the pipeline to intact-cell MALDI-TOF spectra of four *Aeromonas* species in one-against-all tasks; held-out test accuracies ranged from 82.1% to 100% (F1-scores 0.769–1.000), with the smallest test partition (*A. veronii/sobria*, $n = 30$) warranting cautious interpretation. Source code and executable are distributed under an open-source license (repository URL and Zenodo DOI to be inserted before submission), with a versioned archive for long-term citation in line with journal requirements.

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*Corresponding author
Email address: raittz@ufpr.br (Roberto Tadeu Raittz)

1. Introduction

1.1. Mass spectrometry in biomedical research

Mass spectrometry (MS) measures mass-to-charge ratios of ionized analytes and underpins proteomics, metabolomics, and microbial identification workflows [1]. Matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) MS is now widely used for rapid bacterial identification from intact-cell fingerprints [2, 3]. Across domains, comparable computational stages recur: import, preprocessing, feature extraction, and supervised or statistical classification—steps whose quality strongly affects downstream accuracy [4, 5]. The present work focuses on **software** that makes this full chain accessible in one graphical environment rather than on new MS instrumentation or large-scale clinical benchmarking.

1.2. Existing software landscape and its limitations

A variety of software tools address subsets of the MS analysis pipeline. Instrument-specific proprietary solutions—such as workflows tied to Ciphergen ProteinChip technology [6] and MALDI Biotyper (Bruker Daltonics) [7]—provide integrated workflows but restrict users to vendor-specific data formats and algorithms. Platform-independent open-source tools, including MZmine 2 [8], OpenMS, Cromwell, and XCMS [5], offer flexible preprocessing algorithms encompassing smoothing, baseline correction, and peak detection. However, these tools generally do not incorporate GA-based or correlation-driven feature extraction coupled to machine learning classifiers in a unified graphical interface accessible to laboratory biologists without programming expertise.

Feature extraction from MS data poses challenges distinct from general dimensionality reduction: spectral peaks must be aligned across samples with varying m/z resolution, informative peaks must be distinguished from noise and non-discriminant background, and the resulting feature representation must generalize across training and validation partitions to avoid overfitting. Genetic-algorithm (GA)-based approaches to biomarker discovery have shown promise [9, 10], as has the use of cross-validation within the feature extraction loop [10]. Nevertheless, implementations are typically distributed as standalone scripts or require custom integration, presenting a significant barrier to adoption.

1.3. Contribution of this work

We developed MS-ML Studio to address the gap between instrument-specific proprietary suites and fragmented open-source components. Our contributions are as follows:

1. **Unified pipeline**: A single MATLAB graphical application that covers all steps from raw spectral import through MLP-based classification and result visualization.
2. **Novel feature extraction strategies**: Three complementary methods—correlation-based Best Peaks Selection, Watch Points with fuzzy triangular weighting, and Feature Resonance using GA-optimized multi-sinusoidal functions—each guided by K-fold cross-validation to control overfitting.
3. **Platform independence**: The tool handles raw two-column text spectra from any MS platform, without dependence on proprietary data formats.
4. **Illustrative performance**: On the reference intact-cell MALDI-TOF *Aeromonas* case study (Sections 2.1–3), test-set accuracies of 82–100% show that the pipeline can perform competitively under transparent settings; features remain exportable to external classifiers via WEKA.
5. **Open source**: Full source code and compiled executable are publicly available, enabling reproducibility and extension by the community.

Note (branding): **MS-ML Studio** denotes the software described in this manuscript; the version-controlled distribution is maintained under the **MLMS Studio** project name in the public repository (see Availability).

2. Materials and methods

2.1. Data

The following **reference case study** illustrates MS-ML Studio on published intact-cell MALDI-TOF data; it is intended as **proof-of-concept software validation**, not as a standalone clinical or epidemiological study. All mass spectra were acquired on a Bruker MALDI-TOF analyzer from intact bacterial cells of four *Aeromonas* species: *A. hydrophila*, *A. caviae*, *A. veronii/sobria*, and *A. trota*. Sample preparation, bacterial culture, and original spectral acquisition followed the protocol described by Surek *et al.* [11]. A total of 567 spectra were available across the four species (316 for

the *A. veronii/sobria* group). For each of four one-against-all classification experiments, 90% of the available spectra were reserved for feature extraction and classifier training; the remaining 10% were withheld as a blind test set (Table 1).

2.2. MS-ML Studio architecture

MS-ML Studio comprises two components: the MS-ML Library, a set of modular MATLAB functions implementing all algorithmic steps, and the MS-ML User Interface, a MATLAB GUIDE-based graphical application that exposes library functionality through interactive controls. The overall workflow is organized into three phases: (1) preprocessing, (2) feature extraction with cross-validation, and (3) neural network training and classification.

2.3. Preprocessing

2.3.1. Data import

Input spectra are plain-text files with two whitespace-delimited columns: m/z values and relative intensities. This format is instrument-agnostic and can be exported from most MS platforms. There is no built-in converter for proprietary formats such as mzML or mzXML; conversion must be performed with external tools prior to import.

2.3.2. Intensity normalization

Intensity normalization rescales each spectrum's intensity vector to the range $[0, 1]$ using the min–max formula

$$N(x, x_{\max}, x_{\min}) = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}\,, \tag{1}$$

where x is the intensity vector and i indexes each data point.

2.3.3. Spectral reshaping

High-resolution spectra may contain tens of thousands of data points per spectrum. The reshaping operation maps each spectrum onto a user-defined grid of W evenly spaced bins spanning the global m/z range:

$$G(x; W, x_{\max}, x_{\min}) = \mathrm{round}\left(W \cdot N(x; x_{\max}, x_{\min})\right)\,. \tag{2}$$

For each bin $i \in \{1, \ldots, W\}$, the output intensity is the maximum of all input intensities whose rounded grid index equals i, and the output m/z is the mean of the corresponding input m/z values. Reshaping achieves dimensionality reduction and coarse alignment of peak positions across spectra.

2.3.4. Peak alignment

Spectral alignment is applied after reshaping using the Icoshift algorithm [12], which employs gap insertion to maximize the correlation between spectra at user-selected m/z windows. Alignment is optional when internal calibration is sufficient for peak reproducibility across runs.

2.3.5. Peak selection via fuzzy moving average

MS-ML Studio implements a Fuzzy Moving Average (FMA) detector that computes, for each spectral position p, a weighted local average

$$S(x; p, w) = \frac{\sum_{i=p-w}^{p+w} x_i \cdot T(2w+1, i)}{2w+1}, \quad p \in \mathbb{N}^* \mid 1 \leq p-w,\ p+w \leq S\,, \quad (3)$$

where w is the half-width of the triangular window, x is the normalized intensity vector, and $T(L, i)$ is the triangular weighting function defined in Section 2.5. Intensity values exceeding $S(x; p, w)$ at local maxima are retained as candidate peaks.

2.4. Best peaks selection

From the set of candidate peaks identified by FMA, Best Peaks Selection identifies the subset most discriminant for classification. **Correlation-based selection**: for each candidate peak position j, the Pearson correlation between the presence/absence vector of peak j across training spectra and the binary class label vector is computed; peaks whose absolute correlation exceeds a threshold τ are retained. **Probability-based selection**: discriminancy and exclusivity scores are computed from class-conditional peak presence probabilities as in the Markdown manuscript. Both methods produce a ranked list of discriminant spectral positions for subsequent feature extraction.

2.5. Feature extraction

To avoid overfitting, feature extraction algorithms embed K-fold cross-validation within the optimization loop: at each GA iteration, the training set is split into training and validation folds; a configuration is accepted only if the mean of training and validation correlations improves.

2.5.1. Watch Points

Each watch point k centered at position c_k uses a triangular weighting window of half-width L_k. For L odd,

$$w(L,n) = \begin{cases} \dfrac{2n}{L+1} & 1 \le n \le \dfrac{L+1}{2} \\ 2 - \dfrac{2n}{L+1} & \dfrac{L+1}{2} + 1 \le n \le L \end{cases} . \tag{4}$$

For L even,

$$w(L,n) = \begin{cases} \dfrac{2n-1}{L} & 1 \le n \le \dfrac{L}{2} \\ 2 - \dfrac{2n-1}{L} & \dfrac{L}{2} + 1 \le n \le L \end{cases} . \tag{5}$$

The feature value for watch point k on spectrum s is $W_k = \sum w_k \cdot p$, where p is the binary peak-presence vector padded to match w_k. The GA optimizes window half-widths $\{L_k\}$ to maximize the mean absolute Pearson correlation between $\{W_k\}$ and class labels.

2.5.2. Feature Resonance

Feature Resonance encodes peak information as the projection onto a sum of M sinusoidal waves with parameters optimized by the GA:

$$S(x; \mathbf{a}, \mathbf{b}, \mathbf{c}) = \sum_{i=1}^{M} a_i \sin(b_i + c_i x) . \tag{6}$$

The fitness function is the absolute mean Pearson correlation between projection values and class labels. Variants process selected m/z values, best-peak intensities, best-peak m/z values, or watch-point feature vectors.

2.6. Classification

Extracted features from all enabled extractors are concatenated and passed to a multilayer perceptron (MLP) implemented in the MS-ML Library. The MLP is trained using MATLAB's Neural Network Toolbox with Levenberg–Marquardt backpropagation. The training/test split follows the 90%/10% per-species partition. Features can be exported in ARFF format for WEKA [13].

2.7. Performance metrics

Classification performance was evaluated using accuracy, precision, recall, and F1-score [13]:

$$\text{Accuracy} = \frac{TP + TN}{N}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \tag{7}$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{8}$$

3. Results

3.1. Software overview

MS-ML Studio comprises the MS-ML Library and an interactive GUI divided into panels for Preprocessing, Peaks Selection, Feature Extraction, and Classification, with real-time visualization of preprocessing, optimization progress, and classification outputs (confusion matrices, PCA scatter plots).

3.2. Experimental setup (reference case study)

Four one-against-all binary problems were formulated (one per species). Table 1 summarizes dataset partitioning.

Table 1: Dataset partitioning for each one-against-all classification experiment.

Species	Training (n)	Test (n)	Total (n)
A. hydrophila	511	56	567
A. caviae	511	56	567
A. veronii/sobria	286	30	316
A. trota	511	56	567

3.3. Classification performance

Table 2 summarizes metrics on training and test sets. Test F1-scores ranged from 0.769 (*A. veronii/sobria*) to 1.000 (*A. trota*).

Table 2: Classification performance for each *Aeromonas* one-against-all experiment.

Species	Set	Recall	Precision	Accuracy	F1
A. hydrophila	Training	0.889	0.956	0.898	0.921
	Test	0.895	0.971	0.911	0.931
A. caviae	Training	0.902	0.839	0.843	0.869
	Test	0.838	0.886	0.821	0.861
A. veronii/sobria	Training	0.952	0.819	0.944	0.881
	Test	0.833	0.714	0.900	0.769
A. trota	Training	1.000	0.762	0.990	0.865
	Test	1.000	1.000	1.000	1.000

3.4. Feature extraction configurations

Configurations matched the improved manuscript: combined extractors for each species (FR Selected M/Z, FR Best Peaks Intensity/M/Z, Watch Points, multiple correlation thresholds). PCA of extracted features showed clearest separation for *A. trota*, consistent with perfect test performance; overlap remained visible for *A. veronii/sobria* and *A. caviae*, reflecting higher phenotypic similarity within the genus.

4. Discussion

MS-ML Studio achieved test-set accuracies from 82.1% (*A. caviae*) to 100% (*A. trota*) and F1-scores from 0.769 to 1.000, consistent with comparable MALDI-TOF studies [14, 3]. Variability across species aligns with phylogenetic relatedness; the small test set for *A. veronii/sobria* ($n = 30$) limits precision of that point estimate.

The three feature strategies (Best Peaks Selection, Watch Points, Feature Resonance) are complementary; ensemble use generally outperforms a single extractor [10]. K-fold cross-validation inside the GA mitigates overfitting. Compared with tools such as MZmine 2 [8], MS-ML Studio couples preprocessing, discriminant feature learning, and MLP training in one session; WEKA export supports external classifiers.

Limitations include dataset size, plain-text input only (no native mzML/mzXML), MATLAB or MATLAB Runtime dependency, single embedded classifier without integrated significance testing—as detailed in the Markdown source. **Future work** includes standard format parsers, wavelet-based features [10], unsupervised visualization, and multi-class formulations.

5. Conclusion

We presented MS-ML Studio, an open-source MATLAB platform integrating MS preprocessing, three cross-validated feature extractors, and MLP classification in one GUI. On a reference *Aeromonas* MALDI-TOF task, held-out test accuracies of 82–100% and F1-scores of 0.769–1.000 illustrate competitive performance under transparent settings; larger, multi-site studies are required for clinical claims. Modular library design and WEKA export support extension and broader ML workflows.

Availability

Software: Source and executable for MS-ML Studio (MLMS Studio) will be released under an open-source license approved by the institution; replace repository URL and Zenodo DOI placeholders before submission.

Data: The *Aeromonas* MALDI-TOF spectra used in this case study derive from the dataset described by Surek *et al.* [11] and are available upon request from the corresponding author.

Conflict of interest

The authors declare no conflict of interest.

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